

22/9/25 Task 7:- Procedure function and loops: Program  
Using PL/SQL Procedures, functions & loops.

Aim:- To implement PL/SQL procdures, functions  
And loops on number theory and business  
scenarios.

1. Simple PL/SQL program (static input)

```
DECLARE  
    message VARCHAR2(20) := 'Booking closed;  
BEGIN  
    dbms_output.put_line(message);  
END;
```

Output:-

Booking closed

2. Conditional Statement (Dynamic input);

```
DECLARE  
    hid NUMBER(3) := 100;  
BEGIN  
    If (hid = 10) THEN  
        dbms_output.put_line('value of hid is  
                                10');  
    ELSEIF (hid = 20) THEN  
        dbms_output.put_line('value of hid is 20');  
    ELSEIF (hid = 30) THEN  
        dbms_output.put_line('value of hid is 30');  
    ELSE  
        dbms_output.put_line('None of the value's  
                                matching');  
    END IF;  
    dbms_output.put_line('Exact value of hid is'(hid));  
END;
```

Output:-

None of the value is matching  
Exact value of hid is: 100.

### 3. Nested Loops Example:

DECLARE

hid NUMBER(1);

oid NUMBER(1);

BEGIN

LL outer - loop >>

for hid IN 1..3 loop

LL inner - loops >>

for oid IN 1..3 loop

dbms\_output.put\_line ('hid is: || hid ||  
and oid is: || oid ||')

END LOOP inner - loop;

END LOOP outer - loop;

END;

Output:-

hid is: 1 and oid is: 1

hid is: 1 and oid is: 2

hid is: 1 and oid is: 3

hid is: 2 and oid is: 1

hid is: 2 and oid is: 2

hid is: 2 and oid is: 3

hid is: 3 and oid is: 1

hid is: 3 and oid is: 2

hid is: 3 and oid is: 3

### 4. Procedure Example

CREATE OR REPLACE PROCEDURE booking - starts

(L-oid IN NUMBER)

IS

BEGIN.

IF C-id > 200 THEN

dbms\_output.put\_line('No booking available');

ELSE

dbms\_output.put\_line('Booking open');

END IF;

END;

/

Execution-

BEGIN

booking\_status(150);

booking\_status(250);

END;

/

Output:-

Booking open

No Booking available

PL/SQL Procedure for Loops.

Example 1:- Using WHILE Loop with cursor.

Prime check using while loop

CREATE OR REPLACE PROCEDURE print\_prime\_customer IS

CURSOR cust CUR IS

SELECT customer\_id FROM customers;

v\_id NUMBER;

v\_is\_prime BOOLEAN;

v\_i NUMBER;

BEGIN

OPEN cust CUR;

LOOP

FETCH cust CUR INTO v\_id;

```

EXIT THEN CUST - CUX % NOT FOUND;

IF V-id < 2 THEN
    V-is-prime := FALSE;
ELSE
    V-is-prime := TRUE;
    V-i := 2;
    WHILE V-i <= TRUNC(SQRT(V-id)) LOOP
        IF MOD(V-id, V-i) = 0 THEN
            V-is-prime := FALSE;
            EXIT;
        END IF;
        V-i := V-i + 1;
    END LOOP;
END IF;

IF V-is-prime THEN
    DBMS_OUTPUT - PUT-LINE('PRIME CUSTOMER
                                ID: ' || V-id);

END IF;
END LOOP;
CLOSE CUST - CUX;

```

END;

The procedure checks all customer IDs in the Table and prints the prime ones using a WHILE loop.

Example 2:- Using FOR loop for first N prime numbers.

(CREATE OR REPLACE PROCEDURE print-first-n-  
primes (n number) IS

V-num NUMBER := 2;

V-count NUMBER := 0;

V-is-prime BOOLEAN;

BEGIN

WHILE V-count < n LOOP

V-is-prime := TRUE;

FOR i IN 2..TRUNC(SQRT(V-num)) LOOP

IF MOD(V-num, i) = 0 THEN

V-is-prime := FALSE;

EXIT;

END IF;

END LOOP;

IF V-is-prime THEN

DBMS\_OUTPUT.PUT\_LINE('Prime: ' || V-num);

V-count := V-count + 1;

END IF;

V-num := V-num + 1;

END LOOP;

END;

This procedure prints the first N prime numbers  
using a for loop.

BEGIN

Print — first — n — Primes(10);

END;

| VEL TECH                |             |
|-------------------------|-------------|
| EX No.                  | 57          |
| PERFORMANCE (5)         | 5           |
| RESULT AND ANALYSIS (5) | 5           |
| VIVA VOCE (5)           | 5           |
| RECORD (5)              | ✓           |
| TOTAL (20)              | 15          |
| SIGN WITH DATE          | (m) 22/8/20 |

RESULT:- Thus, the procedure function and loops  
programs using PL/SQL procedures, functions &  
loops are executed successfully.